

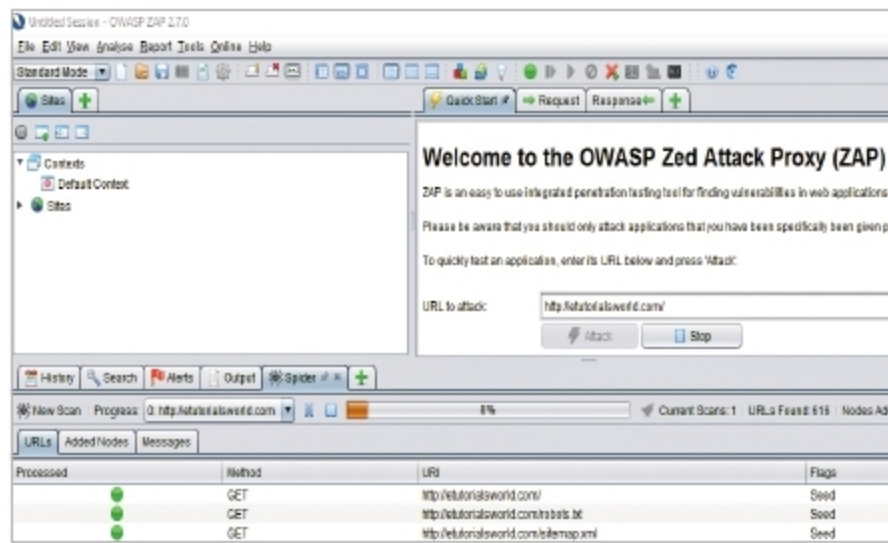


Figure 8: ZAP dashboard

- Next, select *Thread Group* and give it a name. In *Properties*, enter the number of threads, the ramp-up period and the loop count. Right-click on *Thread Group*. Click on *Add > Sampler > HTTP Request*. In *HTTP Request*, provide the server name or IP. Give the port number where your Web server is running.
- Select the *Get method* and provide a path to the load test using Apache JMeter.
- Verify the results.

Security testing

Security testing is useful to find vulnerabilities in the application to protect user information and manage functionality while keeping integrity intact. There are many types of security testing—vulnerability scanning, security scanning, penetration testing, risk assessment, and so on. We will discuss the OWASP ZAP tool in this article.

OWASP Zed Attack Proxy (ZAP) is an open source Web application security scanner. Zed Attack Proxy (ZAP) is an open source penetration testing tool. It is managed under the Open Web Application Security Project (OWASP). Essentially, ZAP works as man-in-the-middle proxy.

Name	OWASP ZAP
Objective	Web application security scanner, penetration testing tool
Licence	Apache Licence
Stable release	2.7.0
Website	https://www.owasp.org/index.php/OWASP_Zed_Attack_Proxy_Project
Supported operating systems	Windows, Linux, Mac
Written in	Java
Support for integration with continuous integration tools	Plugin is available to integrate ZAP with CI tools such as Jenkins

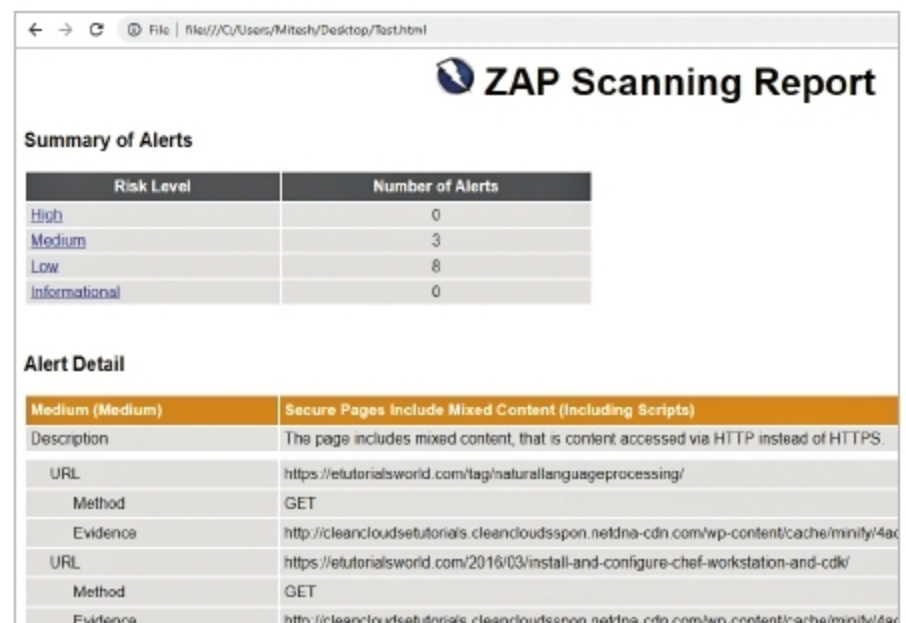


Figure 9: ZAP HTML report

Go to <https://github.com/zaproxy/zaproxy/wiki/Downloads>. Enter the URL to quickly test an application and click on *Attack!*

To generate a report after a quick test, click on the *Report* menu and select *Generate HTML Report*. Save the report.

Automation in testing is a good practice to adopt in the beginning of an application's life cycle. Unit testing can be easily done using modern IDEs. Open source function testing, even with 40-50 per cent coverage, is very helpful. **END** 🐧

References

- [1] Jenkins Essentials (Second edition): Setting the stage for a DevOps culture; <https://www.amazon.com/gp/product/B073PGCKHY>
- [2] Jenkins 2.x Continuous Integration Cookbook; <https://www.amazon.com/gp/product/B0751HZ96H>
- [3] Implementing DevOps with Microsoft Azure; <https://www.amazon.com/gp/product/1787127028>
- [4] DevOps for Web Development; <https://www.amazon.com/gp/product/B01HY3TC2M>
- [5] DevOps Bootcamp: The fastest way to learn DevOps; <https://www.amazon.com/gp/product/B07259Y9M8>
- [6] Articles related to Selenium; <https://etutorialsworld.com/category/selenium/>
- [7] Articles related to Apache JMeter; <https://etutorialsworld.com/category/jmeter/>
- [8] Articles related to SMAC; <https://etutorialsworld.com/category/smac/>

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